MATH1141: Higher Mathematics 1B (Algebra)

L. Cheung

August 13, 2026

Contents

Chapter 1

Functions of several variables

1.1 Motivations, sketching surface, and partial derivatives

1.1.1 Function of two variables

We will consider functions of the type

$$z = F(x, y), \quad (x, y) \in D$$

where $D \subseteq \mathbb{R}^2$ is the domain of F and F is real-valued. We write $F : D \subseteq \mathbb{R}^2 \to \mathbb{R}$.

For example:

- $F(x, y) = 2x + y^6$ and $D = \mathbb{R}^2$
- $F(x, y) = xe^y$ and $D = \mathbb{R}^2$

Note: if $D = \mathbb{R}^2$, sometimes the domain is not explicitly written.

Geometric representation

The function of one variable: $y = f(x), \quad x \in [a, b]$ can be represented as a curve in $\mathbb{R}^2$.

Similarly, a function of two variables: $z = F(x, y), \quad (x, y) \in D$ can be described as a surface in $\mathbb{R}^3$

1.1.2 Sketching and visualising surfaces

Consider a surface defined by

$$z = F(x, y).$$

Conventionally, z is the distance above or below the x, y-plane.

A **level curve** of a function, $F : \mathbb{R}^2 \to \mathbb{R}$, is a curve in $\mathbb{R}^2$ defined by $F(x, y) = C$, where C is a constant. This can be visualised by cutting a surface with a plane. Typically this plane is parallel to the xy-plane

Example

Sketch the surface defined by

$$z = F(x, y) = x^2 + y^2$$

1. Sketch some level curves for different values C.
   - Set $z = C = 0 \Rightarrow \quad x^2 + y^2 = 0 \Rightarrow$ Singular point at origin
   - Set $z = C = 1 \Rightarrow \quad x^2 + y^2 = 1 \Rightarrow$ Circle centred at 0 with radius 1
   - Set $z = C = 4 \Rightarrow \quad x^2 + y^2 = 4 \Rightarrow$ Circle centred at 0 with radius 2
   - Set $z = C = 8 \Rightarrow \quad x^2 + y^2 = 8 \Rightarrow$ Circle centred at 0 with radius $2\sqrt{2}$
2. Sketch the curves of intersection between the surface $z = x^2 + y^2$ and the yz-plane (set $x = 0$) or xz-plane (set $y = 0$).
   - In this example, both the intersections form parabolas.

Example

Sketch the surface defined by $x^2 + y^2 - z^2 = 1$.

1. Level curves
   - level curves with $z = 0$ is a circle with radius 1
   - level curves with $z = \pm 1$ are circles with radius $\sqrt{2}$
   - level curves with $z = \pm 2$ are circles with radius $\sqrt{5}$
2. Intersections between the surface and the yz-plane gives $y^2 - z^2 = 1$.
3. These pieces of information can be put together to form the surface

1.1.3 Partial differentiation

Informally: differentiate with respect to one of the variables by holding the other variable fixed.

Definition

If $F : \mathbb{R}^2 \to \mathbb{R}$ is a function of two variables x and y, then the partial derivatives of $F(x, y)$ with respect to x and y at the point (x_0, y_0) are defined by

$$\frac{\partial F}{\partial x}(x_0, y_0) = \lim_{h \to 0} \frac{F(x_0 + h, y_0) - F(x_0, y_0)}{h},$$

and

$$\frac{\partial F}{\partial y}(x_0, y_0) = \lim_{h \to 0} \frac{F(x_0, y_0 + h) - F(x_0, y_0)}{h},$$

wherever these limits exist.

If a function F has partial derivatives with respect to x and y, then:

- The partial derivative with respect to x may be denoted by

$$\frac{\partial F}{\partial x} \equiv F_x \equiv D_1 F.$$

- The partial derivative with respect to y may be denoted by

$$\frac{\partial F}{\partial y} \equiv F_y \equiv D_2 F.$$

The rules for partial derivatives are similar to that of normal derivatives.

Example

Consider

$$F(x, y) = e^{y^2} \cos(x^2 y) + e^{x^2 y} \sin(y).$$

Compute the partial derivatives of $\frac{\partial F}{\partial x}$ and $\frac{\partial F}{\partial y}$.

$$\begin{aligned} \frac{\partial F}{\partial x} &= \frac{\partial}{\partial x}\left[e^{y^2} \cos(x^2 y)\right] + \frac{\partial}{\partial x}\left[x^{y^2} \sin(y)\right] \\ &= e^{y^2}\left[-\sin(x^2 y)(2xy)\right] + \sin(y) e^{y^2}(2xy) \end{aligned}$$

1.1.4 Geometrical Interpretation

Recall that $\frac{\partial F}{\partial x}(x_0, y_0) = \lim_{h \to 0} \frac{F(x_0 + h, y_0) - F(x_0, y_0)}{h}$.

$\frac{\partial F}{\partial x}(x_0, y_0)$ is the slope of the tangent to the cross section at (x_0, y_0) when the surface $z = F(x, y)$ is intersected with the plane $y = y_0$.

This can be applied similarly to the derivative $\frac{\partial F}{\partial y}$

1.1.5 Second Order Partial Derivatives

A second order partial derivative is a partial derivative of a first order partial derivative.

$$\frac{\partial^2 F}{\partial x} \equiv \frac{\partial}{\partial x}(\frac{\partial F}{\partial x}) \equiv F_{xx} \equiv D_1 D_1 F$$

$$\frac{\partial^2 F}{\partial y} \equiv \frac{\partial}{\partial y}(\frac{\partial F}{\partial y}) \equiv F_{yy} \equiv D_2 D_2 F$$

$$\frac{\partial^2 F}{\partial x \partial y} \equiv \frac{\partial}{\partial x}(\frac{\partial F}{\partial y}) \equiv F_{yx} \equiv D_1 D_2 F$$

$$\frac{\partial^2 F}{\partial y \partial x} \equiv \frac{\partial}{\partial y}(\frac{\partial F}{\partial x}) \equiv F_{xy} \equiv D_2 D_1 F$$

The Mixed Derivative Theorem

Suppose that F is a function of two variables. If F and all its first and second order partial derivatives are continuous, then

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x}$$

Definition

Consider a function G on $\mathbb{R}^2$ with variables x and y. The function G is continuous at the point $(x_0, y_0) \in \mathbb{R}^2$ if

$$\lim_{\substack{x \to x_0 \\ y \to y_0}} G(x, y) = F(x_0, y_0).$$

- Finite sums/products of continuous functions are continuous, and compositions of continuous functions are continuous
- The mixed derivative theorem tells us that for a function which can be written as a finite sum/product of a polynomials, sine, cosine, exponential functions, all derivatives of all orders are continuous, and so we get the equality of all mixed derivatives.
- The existence of partial derivatives at a point does not imply continuity at a point

Example

Consider the function

$$F(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{for } (x, y) \neq (0, 0) \\ 0 & \text{for } (x, y) = (0, 0) \end{cases}.$$

We see that $\frac{\partial F}{\partial x}(0, 0) = \frac{\partial F}{\partial y}(0, 0) = 0$, but $F(x, y)$ is not continuous at $(0, 0)$.

Chapter 2

Tangent planes, normal vectors, and error estimation

2.1 Tangent Lines and Normals

Suppose that (x_0, y_0) is a point that lies on the curve $y = f(x)$. If the curve has a tangent line at (x_0, y_0):

- The tangent line has the equation

$$y = y_0 + f'(x_0)(x - x_0).$$

- A normal vector to the line at (x_0, y_0) is

$$\begin{pmatrix} f'(x_0) \\ -1 \end{pmatrix}.$$

Suppose that (x_0, y_0, z_0) is a point that lies on the curve $z = F(x, y)$. If the curve has a tangent line at (x_0, y_0, z_0):

- The tangent plane has the equation

$$z = z_0 + F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0).$$

- A normal vector to the surface at (x_0, y_0, z_0) is

$$\begin{pmatrix} F_x(x_0, y_0) \\ F_y(x_0, y_0) \\ -1 \end{pmatrix}.$$

Consider vectors along each of the tangent lines

- A vector along the tangent line to the curve $z = F(x, y_0)$ is

$$\begin{pmatrix} 1 \\ 0 \\ F_x(x_0, y_0) \end{pmatrix}.$$

- A vector along the tangent line to the curve $z = F(x_0, y)$ is

$$\begin{pmatrix} 0 \\ 1 \\ F_y(x_0, y_0) \end{pmatrix}.$$

A normal vector to the tangent plane is given by the cross product

$$\begin{pmatrix} 0 \\ 1 \\ F_y(x_0, y_0) \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ F_x(x_0, y_0) \end{pmatrix} = \begin{pmatrix} F_x(x_0, y_0) \\ F_y(x_0, y_0) \\ -1 \end{pmatrix}.$$

The point normal form of the tangent plane

$$\begin{pmatrix} F_x(x_0, y_0) \\ F_y(x_0, y_0) \\ -1 \end{pmatrix} \cdot \begin{pmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{pmatrix} = 0$$

Rearranging terms gives

$$z = z_0 + F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0).$$

Example

Consider the surface S defined by $z = F(x, y) = -\frac{x^2}{4} - y^2$. Find the tangent plane of S at $(x_0, y_0, z_0) = (2, -1, -2)$ and the associated normal vector of this tangent plane.

$$F_x = \frac{\partial F}{\partial x} = -\frac{x}{2}$$

$$F_y = \frac{\partial F}{\partial y} = -2y$$

At $(x_0, y_0) = (2, -1)$,

$$F_x(2, -1) = F_x(x_0, y_0) = -1$$

$$F_y(2, -1) = 2$$

The normal vector can be given as

$$\begin{pmatrix} F_x(2, -1) \\ F_y(2, -1) \\ -1 \end{pmatrix} = \begin{pmatrix} -1, 2, -2 \end{pmatrix}$$

The tangent plane is

$$z - z_0 = F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0)$$

$$z = -x + 2y + 2$$

2.2 Total Differential Approximation

If $F = F(x, y)$, then near a given point (x_0, y_0),

$$F(x, y) \approx F(x_0, y_0) + F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0).$$

Define

$$\Delta F = F(x, y) = F(x_0, y_0)$$

$\Delta F = F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0)$ is the total differential approximation. Geometrically this approximates points on the surface $z = F(x, y)$ near (x_0, y_0) by points on the tangent plane near this point.

If a function F is smooth enough then the tangent plane at the point (x_0, y_0) will provide a good approximation to F for points near to (x_0, y_0). Smooth means that the surface has no corners, sharp peaks, or folds.

$$\begin{aligned} \text{Error} &= |\Delta F| \\ &\approx \left| \frac{\partial F}{\partial x} \Delta x + \frac{\partial F}{\partial y} \Delta y \right| \\ &\leq \left| \frac{\partial F}{\partial x} \right| |\Delta x| + \left| \frac{\partial F}{\partial y} \right| |\Delta y| \end{aligned}$$

Example

A right circular cone is measured to have a height of 10 cm and a base radius of 30 cm. Estimate an upper bound of the absolute error in the volume if the maximum absolute error in each measurement is 0.1 cm.

$$\begin{cases} \frac{\partial V}{\partial r} = \frac{2}{3}\pi r h \\ \frac{\partial V}{\partial h} = \frac{1}{3}\pi r^2 \end{cases}$$

$$V = \frac{1}{3}(\pi r^2)h = F(r, h)$$

$$|\Delta V| \leq \left| \frac{\partial V}{\partial r}(r_0, h_0) \right| |\Delta r| + \left| \frac{\partial V}{\partial h}(r_0, h_0) \right| |\Delta h|$$

Since $|\Delta r| \leq 0.1$ and $|\Delta j| \leq 0.1$

$$\begin{aligned} |\Delta V| &\leq \left| \frac{2}{3}\pi \cdot 30 \cdot 10 \right| (0.1) + \left| \frac{1}{3}\pi 30^2 \right| (0.1) \\ &= 50\pi \end{aligned}$$

Chapter 3

Chain Rules and Functions with three (or more) variables

3.1 Chain Rules

If $F = F(x, y)$ and $x = x(t)$, $y = y(t)$, then

$$\frac{dF}{dt} = \frac{\partial F}{\partial x}\frac{dx}{dt} + \frac{\partial F}{\partial y}\frac{dy}{dt}$$

Chapter 4

Trigonometric Integrals and Reduction Formula

Example

Evaluate $\int \cos^4 x \sin^3 x dx$.

$$\begin{aligned}
\int \cos^4 x \sin^3 x dx &= \int \cos^4 x \sin^2 x \sin x dx \\
&= \int \cos^4 x (1 - \cos^2 x) \frac{d(-\cos x)}{dx} dx \\
&= \int u^4 (1 - u^2)(-1) du \\
&= \int (u^6 - u^4) du \\
&= \frac{u^7}{7} - \frac{u^5}{5} + c \\
&= \frac{(\cos x)^7}{7} - \frac{(\cos x)^5}{5} + c
\end{aligned}$$

4.1 Irrationality of π

- Proof by contradiction, assuming π is rational and showing that it leads to a contradiction.
- Show that if π is rational, then a certain integral must be an integer. We will also show that the integral must lie on the interval $(0, 1)$, hence the contradiction.

For a non-negative integer n, and a positive integer q, define:

$$I_n = \frac{q^{2n}}{n!} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\pi^2}{4} - x^2\right)^n \cos(x) dx.$$

Using integration by parts (twice), one can show, if $n \geq 2$,

$$I_n = (4n - 2)q^2 I_{n-1} - q^4 \pi^2 I_{n-2},$$

with $I_1 = 4q^2$, and $I_0 = 2$.

Assuming that $\pi = \frac{p}{q}$, where p and q are positive integers, show that I_n is an integer for all integers $n \geq 0$.

We know that I_0 and I_1 are integers. Suppose I_{k-2} and I_{k-1} are integers whenever $k \geq 2$. By the assumption that $\pi = \frac{p}{q}$, using previous identity on I_k we have

$$\begin{aligned} I_k &= (4k - 2)q^2 I_{k-1} - q^4 \pi^2 I_{k-2} \\ &= (4k - 2)q^2 I_{k-1} - q^4 (\frac{p}{q})^2 I_{k-2} \\ &= (4k - 2)q^2 I_{k-1} - q^2 p^2 I_{k-2}, \end{aligned}$$

which is also an integer. Hence I_n is an integer for all positive integer n.

Using the previous assumption about π and considering the inequality,

$$0 < \left(\frac{\pi^2}{4} - x^2\right)^n \cos(x) \leq \left(\frac{\pi^2}{4}\right)^n$$

for $n \geq 0$ and $-\frac{\pi}{2} < x < \frac{\pi}{2}$, we see that

$$0 < I_n < \frac{q^{2n}}{n!} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\pi^2}{4}\right)^n dx = \frac{p}{q} \frac{\left(\frac{p^2}{4}\right)^n}{n!}.$$

Using the fact that

$$\lim_{n\to\infty} \frac{a^n}{n!} = 0,$$

for any real number a with $a \geq 0$, we see that $0 < I_n < 1$ for all large n which contradicts the fact that I_n is an integer for all n.

Chapter 5

Trigonometric Substitution, Partial Fractions, and Other Substitution

Example

Find $I = \int_0^4 x^2\sqrt{16 - x^2}dx$.
Let $x = 4\sin\theta, 0 \leq \theta \leq \frac{\pi}{2}$.

$$\begin{aligned}
I &= \int_0^{\frac{\pi}{2}} 16\sin^2\theta \cdot \sqrt{16 - 16\sin^2\theta} \cdot (4\cos\theta) \cdot d\theta \\
&= \int_0^{\frac{\pi}{2}} 16\sin^2\theta \cdot (4\cos\theta) \cdot (4\cos\theta) \cdot d\theta \\
&= 16^2 \cdot \int_0^{\frac{\pi}{2}} \sin^2\theta \cdot \cos^2\theta \cdot d\theta \\
&= 16^2 \cdot \int_0^{\frac{\pi}{2}} \frac{\sin^2(2\theta)}{4} \cdot d\theta \\
&= 16^2 \cdot \int_0^{\frac{\pi}{2}} \left[\frac{1 - \cos(4\theta)}{8}\right] \cdot d\theta \\
&= 16\pi.
\end{aligned}$$

The integral of the form $\int \sqrt{a^2 - x^2}$ is geometrically useful in finding the area enclosed by an ellipse, ie. $\frac{x^2}{a} + \frac{y^2}{b} = 1$ has an area

$$\begin{aligned}
A &= 4 \cdot \int_0^a \frac{b}{a}\sqrt{a^2 - x^2} \cdot dx \\
&= \pi ab.
\end{aligned}$$

5.1 Rational functions and partial fractions

- A function f is a rational function if it is of the form $f(x) = \frac{P(x)}{Q(x)}$ where $P(x)$ and $Q(x)$ are polynomials.
- A rational function f is said to be proper if the degree of the numerator $P(x)$ is less than the dress of the denominator $Q(x)$.
- An improper rational function can be written as the sum of a polynomial and a proper rational function:

$$f(x) = \frac{P(x)}{Q(x)} = p(x) + \frac{R(x)}{Q(x)}.$$

Example

Find $\int \frac{2x+1}{x^2+4x+5} \cdot dx$

$$\begin{aligned} I &= \int \frac{2x+1}{x^2+4x+5} \cdot dx \\ &= \int \left[\frac{2x+4}{x^2+4x+5} - \frac{3}{x^2+4x+5} \right] \cdot dx \\ &= \ln(x^2+4x+5) - 3\arctan(x+2) + C. \end{aligned}$$

Example

Find $\int \frac{2x^2+2x-6}{(x-1)(x+1)(x-2)}$.

Use partial fraction:

$$\frac{2x^2+2x-6}{(x-1)(x+1)(x-2)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x-2}$$

$$\dots$$

$$= \begin{cases} A = 1 \\ B = -1 \\ C = 2 \end{cases}$$

therefore

$$\begin{aligned} \int \frac{2x^2+2x-6}{(x-1)(x+1)(x-2)} &= \int \left[\frac{1}{x-1} + \frac{-1}{x+1} + \frac{2}{x-2} \right] dx \\ &= \ln|x-1| - \ln|x+1| + 2\ln|x-2| + D. \end{aligned}$$

5.2 Other substitutions

Example

Find $\int \frac{dx}{1+x^{\frac{1}{4}}}$.

Let $u = x^{\frac{1}{4}}$, $x = u^4$, and therefore $dx = 4u^3 du$.

$$
\begin{aligned}
I = \int \frac{dx}{1 + x^{\frac{1}{4}}} &= \int \frac{4u^3 du}{1 + u} \\
&= 4 \int \left[(u^2 - u + 1) + \frac{-1}{1 + u} \right] du \\
&= 4 \left[\frac{u^3}{3} - \frac{u^2}{2} + (-1) \ln |1 + u| \right] + C \\
&= 4 \left[\frac{(x^{\frac{1}{4}3})}{3} - \frac{(x^{\frac{1}{4}})^2}{2} - \ln(1 + x^{\frac{1}{4}}) \right] + C
\end{aligned}
$$

Example

Find $\int \frac{dx}{\sqrt{e^{2x}-1}}$.

Let $u = \sqrt{e^{2x} - 1}$, $u^2 = e^{2x} - 1$, $u^2 + 1 = e^{2x}$, therefore $2udu = 2e^{2x}dx$, $2u\frac{du}{dx} = 2(u^2 + 1)$

$$
\begin{aligned}
I &= \int \frac{dx}{\sqrt{e^{2x} - 1}} \\
&= \int \left[\frac{1}{u} \cdot \frac{u}{u^2 + 1} \right] du \\
&= \arctan(u) + C = \arctan(\sqrt{e^{2x} - 1}) + C
\end{aligned}
$$

5.3 Weierstrass substitution

- Weierstrass substitution is useful for integrals with fractional expressions in terms of $\sin x$ and/or $\cos x$
- Weierstrass substitution is

$$t = \tan \frac{x}{2}$$

- Recall that $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $dx = \frac{2dt}{1+t^2}$

Example

Calculate

$$I = \int \frac{dx}{1 + \sin x + \cos x}.$$

$$\begin{aligned} I = \int \frac{dx}{1 + \sin x + \cos x} &= \int \frac{\frac{2}{1+t^2}dt}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \\ &= \int \frac{2}{(1 + t^2) + 2t + (1 - t^2)}dt \\ &= \int \frac{1}{1 + t}dt \\ &= \ln|1 + \tan(\frac{x}{2})| + C \end{aligned}$$

Example

Calculate

$$I = \int_1^2 \frac{x^2 - 1}{x^2 + 1} \frac{1}{\sqrt{1 + x^4}}dx.$$

Hint: Try the substitution $u = x + \frac{1}{x}$ or $u = \sqrt{x^2 + \frac{1}{x^2}}$

Let $u = x + \frac{1}{x}$, $u^2 = (x + \frac{1}{x})^2$

$$I = \int_1^2 \frac{x - \frac{1}{x}}{x + \frac{1}{x}} \frac{1}{\sqrt{x^2(\frac{1}{x^2} + x^2)}}dx$$

$$= \int_2^{\frac{5}{2}} \frac{du}{u} \frac{1}{\sqrt{u^2 - 2}}$$

Let $u = \sqrt{2} \sec \theta$, therefore $du = \sqrt{2} \sec \theta \tan \theta$

$$= \int_{\sec^{-1}(\sqrt{2})}^{\sec^{-1}(\frac{5}{2\sqrt{3}})} \frac{\sqrt{2} \sec \theta \tan \theta}{\sqrt{2} \sec \theta} \frac{1}{\sqrt{2} \tan \theta}d\theta$$

$$= \frac{1}{\sqrt{2}} \left[\sec^{-1}\left(\frac{5}{2\sqrt{2}}\right) - \sec^{-1}(\sqrt{2})\right]$$

Chapter 6

First Order Ordinary Differential Equations

A differential equation is an equation that relates an unknown function with one or more of its derivatives. Eg.

$$m\frac{d^x}{dt^2} = -kx$$

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2}$$

Ordinary differential equations do not have any partial derivatives.

Definition

An ordinary differential equation or ODE is an equation involving one independent real variable, one dependent real variable, and finitely many of its derivatives. Thus it is of the form

$$F(x, u(x), u'(x), \ldots, u^{(n)}(x)) = 0.$$

The order of an ODE is the order of the highest derivative present. Eg,

$$\frac{dy}{dx} = y + x^2, \quad \text{is a 1st order ODE}$$

$$y'' + 2xy = 0, \quad \text{is a 2nd order ODE}$$

An nth order ODE is said to be linear if it is a linear combination of the first n derivatives of the dependent function.

6.0.1 Implicit solution

Sometimes a solution to an ODE is given in implicit form.

Example

Show that the function y given by

$$y + \arctan y = \frac{x^2}{2}$$

is a solution to

$$y' = \frac{x(1 + y^2)}{2 + y^2}$$

Differentiate with respect to x,

$$\left(\frac{2 + y^2}{1 + y^2}\right)\frac{dy}{dx} = \frac{dy}{dx} + \frac{2}{1 + y^2}\frac{dy}{dx} = x$$

$$\frac{dy}{dx} = x\left(\frac{1 + y^2}{2 + y^2}\right)$$

6.0.2 Separable ODEs

A separable equation has the form

$$\frac{dy}{dx} = g(x)H(y),$$

ie. we can separate x and y.

Solving

- Case 1: If there exists y_0 such that $H(y_0) - 0$, then $y \equiv y_0$ is a trivial solution
- Case 2: In the remaining case,
  - Letting $h(y) = \frac{1}{H(y)}$, we can rewrite

    $$h(y)\frac{dy}{dx} = g(x).$$

  - Integrate both sides with respect to x:

    $$\int h(y)\frac{dy}{dx}dx = \int g(x)dx,$$

    or

    $$\int h(y)dy = \int g(x)dx.$$

Example

Solve $y' = x\sqrt{1 - y^2}$.

$$\frac{dy}{dx} = x\sqrt{1 - y^2}$$

$$\frac{dy}{dx}\frac{1}{\sqrt{1 - y^2}} = x$$

$$\int \frac{1}{\sqrt{1 - y^2}}dy = \int xdx$$

$$y = \sin\left(\frac{x^2}{2} + C\right)$$

6.0.3 Linear ODEs

A first order linear ODE is an ODE of the form

$$\frac{dy}{dx} + f(x)y = g(x),$$

where f and g are given functions.

Solving

- Calculate the integrating factor $h(x) = e^{\int f(x)dx}$ (ignoring the constant of integration)
- Multiply the given equation by $h(x)$ and use the product rule for differentiation for the left-hand side to rewrite the resulting equation as

$$\frac{d}{dx}(h(x)y) = g(x)h(x).$$

Example

Solve $\frac{dy}{dx} + 3y = e^{-x}$.

Integrating factor: $e^{\int 3dx} = e^{3x}$.
Multiply e^{3x} on both sides

$$e^{3x}\frac{dy}{dx} + 3e^{3x}y = e^{2x}$$

$$\frac{d}{dx}\left[e^{3x} \cdot y\right] =$$

$$\int \frac{d}{dx}\left[e^{3x} \cdot y\right] dx = \int e^{2x} dx$$

$$y(x) = e^{-3x}\left[\frac{e^{2x}}{2} + C\right]$$

Many first order ODEs can be written in the form

$$M(x, y) + N(x, y)\frac{dx}{dy} = 0.$$

If the left-hand hide can be written as $\frac{d}{dx}F(x, y(x))$ then the equation has a solution in implicit form $F(x, y) = C$.

Given a function $F(x, y)$ and any constant C, consider the expression

$$F(x, y) = C,$$

which defines y implicitly in terms of x. By differentiating with respect to x we obtain

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dx}{dy} = 0.$$

If

$$M(x, y) = M = \frac{\partial F}{\partial x} \quad \text{and} \quad N(x, y) = N = \frac{\partial F}{\partial y},$$

then reversing the above steps, we could conclude that a solution is $F(x, y) = C$.

Example

Solve $(2x + y + 1)dx + (2y + x + 1)dy = 0$.

Let $M(x, y) = 2x + y + 1$ and $N(x, y) = 2y + x + 1$. Then

$$\frac{\partial M}{\partial y} = 1 = \frac{\partial N}{\partial x}$$

so the equation is exact. Hence there exists a function $F(x, y)$ satisfying

$$\frac{\partial F}{\partial x} = M(x, y) = 2x + y + 1$$

$$\frac{\partial F}{\partial y} = N(x, y) = 2y + x + 1.$$

Substitutions

If the ODE is not linear, separable or exact, then it may be possible to use a change of variables so that the ODE in the new variables is linear, separable or exact.

Consider ordinary differential equation of the form

$$\frac{dy}{dx} = g(\frac{y}{x}).$$

Setting $y(x) = xv(x)$ will lead to a separable ODE

Example

Solve $2y\frac{dy}{dx} + \frac{x^2+y^2}{x} = 0, \quad x > 0$.

$$\begin{aligned} \frac{dy}{dx} &= -\left(\frac{x^2 + y^2}{x}\right) \cdot \frac{1}{2y} \\ &= -\left(\frac{x^2 + y^2}{2xy}\right) \\ &= -\frac{x}{2y} - \frac{y}{2x} \\ &= -\frac{1}{2}(\frac{x}{y} + \frac{y}{x}) \end{aligned}$$

Let $v = \frac{y}{x}$

$$= g(v), \quad \text{where } g(v) = -\frac{1}{2}(\frac{1}{v} + v)$$

6.1 Second Order Linear ODEs

- Homogeneous Equation (HE)

$$y'' + ay' + by = 0$$

  where $y = y(x)$ is an unknown function, a and b are constants.

- Non-homogeneous Equation (NE)

$$y'' + ay' + by = f(x)$$

  where $y = y(x)$ is an unknown function, a and b are constants, and $f(x)$ is a given function.

- If $b = 0$, then it can be reduced to a first order ODE

For the homogeneous case, if y_1 and y_2 are two solutions of (HE) then so is $Ay_1 + By_2$ where A and B are constants.

(HE) has two linearly independent solutions, and every solution is a linear combination of these solutions.

6.1.1 Finding linearly independent solutions of (HE)

1. Rewrite (HE): $y'' + ay' + by = 0$
2. Look for a solution that does not change too much when differentiated so that the terms on the left-hand side cancel each other out to give zero.
3. Try $y = e^{\lambda x}$ where λ is a constant to be determined.
4. Differentiate and substitute into (HE):

$$e^{\lambda x}(\lambda^2 + a\lambda + b) = 0,$$

   which gives, after dividing by $e^{\lambda x}$

$$\lambda^2 + a\lambda + b = 0.$$

5. So, $y = e^{\lambda x}$ is a solution to (HE) if and only if λ is a solution. This equation is called the characteristic equation of (HE).

Solution Cases

- Case 1: If the characteristic equation has two distinct real solutions λ_1 and λ_2, then

$$y_1 = e^{\lambda_1 x} \quad \text{and} \quad y_2 = e^{\lambda_2 x}$$

  and the general solution to (HE) is

$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}, \quad A, B \in \mathbb{R}$$

- Case 2: If the characteristic equation has one double root λ, then $y_1 = e^{\lambda_1 x}$. It turns out that $y_2 = xe^{\lambda_1 x}$ is another solution to (HE).

  Verify $y_2 = xe^{\lambda_1 x}$ is a solution for $y'' + ay' + by = 0$ in this case.

$$\begin{aligned} y_2'(x) &= e^{\lambda_1 x} + x\lambda_1 e^{\lambda_1 x} \\ y_2''(x) &= \lambda_1 e^{\lambda_1 x} + \lambda_1 e^{\lambda_1 x} + x\lambda_1^2 e^{\lambda_1 x} \\ &= 2\lambda_1 e^{\lambda_1 x} + x\lambda_1^2 e^{\lambda_1 x} \end{aligned}$$

  Substituting yields

$$\begin{aligned} y_2''(x) + ay_2'(x) + by_2(x) &= 2\lambda_1 e^{\lambda_1 x} + x\lambda_1^2 e^{\lambda_1 x} + a\left[e^{\lambda_1 x} + x\lambda_1 e^{\lambda_1 x}\right] + bxe^{\lambda_1 x} \\ &= xe^{\lambda_1 x}\left[\lambda_1^2 + a\lambda_1 + b\right] + e^{\lambda_1 x}\left[2\lambda_1 + a\right] \\ &= 0. \end{aligned}$$

  Hence, the general solution to (HE) is

$$y = Ae^{\lambda x} + Bxe^{\lambda x}, \quad A, B \in \mathbb{R}.$$

- Case 3: If the characteristic equation has two complex roots $\lambda_1 = \alpha + \beta i$ and $\lambda_2 = \alpha - \beta i$, then

$$y_1 = e^{(\alpha+\beta i)x} = e^{\alpha x}e^{i\beta x} \quad \text{and} \quad y_2 = e^{(\alpha-\beta i)x} = e^{\alpha x}e^{-i\beta x}.$$

  The general solution to (HE) is

$$y = e^{\alpha x}(A\cos\beta x + B\sin\beta x), \quad A, B \in \mathbb{R}.$$

Example

Solve the IVP

$$y'' + 6y' + 9y = 0, \quad y(0) = 0, \quad y'(0) = 5.$$

Characteristic equation

$$1 \cdot \lambda^2 + 6\lambda + 9 = 0$$

$$(\lambda + 3)^2 = 0, \quad \lambda = -3 \quad \text{(repeated roots).}$$

So,

$$y(x) = Ae^{-3x} + Bxe^{-3x}$$

$$0 = y(0) = A \cdot 1 + B \cdot 0 \Rightarrow A = 0$$

$$y(x) = Bxe^{-3x}$$

$$y'(x) = Be^{-3x} + Bx(-3)e^{-3x}$$

$$5 = y(0) = B \cdot e^0 = B$$

So, $y(x) = 5xe^{-3x}$

6.1.2 The non-homogeneous case

If y_1 and y_2 are solutions of (NE) then $y_1 - y_2$ is a solution of (NE).

The general solution of (NE) is given by $y = y_h + y_p$ where y_h is the general solution of (HE) and y_p is a particular solution of (NE).

Example

Solve $y'' - 5y' + 6y = 6x + 7$.

The solution will have the form $y(x) = y_h(x) + y_p(x)$, where $y_h(x)$ solves the homogeneous equation $y'' - 5y' + 6y = 0$.

Characteristic equation for (HE) is $\lambda^2 - 5\lambda + 6 = 0 \Rightarrow \lambda = 2, \lambda = 3$. Therefore $y_h(x) = Ae^{2x} + Be^{3x}$.

To find $y_p(x)$

$$y_p(x) = ax + b$$

$$y_p'(x) = a, \quad y_p''(x) = 0$$

$$\begin{aligned} 6x + 7 &= y_p''(x) - 5y_p'(x) + 6y_p(x) \\ &= 0 - 5a + 6(ax + b). \end{aligned}$$

Therefore, $y_p(x) = x + 2$. So, $y(x) = Ae^{2x} + Be^{3x} + (x + 2)$, where A, B are real constants.

Note:

- If any term of the guess for y_p is part of y_h, then multiply that term by x
- If the new guess is still part of y_h, multiply by x again

Example

Solve $y'' + 4y' + 4y = e^{-2x}$.

Solving the homogeneous equation $y'' + 4y' + 4y = 0$, $\lambda^2 + 4\lambda + 4 = 0 \Rightarrow \lambda = -2$, hence $y_h(x) = Ae^{-2x} + Bxe^{-2x}$

For $y_p(x)$, try

$$\left[C \cdot e^{-2x}\right] \cdot x^2.$$

$$y_p'(x) = C(-2)e^{-2x} \cdot x^2 + Ce^{-2x} \cdot (2x)$$

$$\begin{aligned} y_p''(x) &= 4C \cdot e^{-2x} \cdot x^2 + C \cdot (-2)e^{-2x} \cdot 2x + (-2)Ce^{-2x} \cdot (2x) + 2Ce^{-2x} \\ &= 2Ce^{-2x} - 8Cxe^{-2x} + 4Ce^{-2x} \cdot x^2 \end{aligned}$$

Substitute to find a solution for y,

$$\begin{aligned} e^{-2x} &= y_p''(x) + 4y_p'(x) + 4y_p(x) \\ &= 2Ce^{-2x} - 8Cxe^{-2x} + 4Ce^{-2x} \cdot x^2 + 4\left[C(-2)e^{-2x} \cdot x^2 + Ce^{-2x} \cdot (2x)\right] + 4Ce^{-2x} \cdot x^2 \end{aligned}$$

Therefore, $C = \frac{1}{2}$.

Chapter 7

Taylor Polynomials

Taylor polynomials allow the approximation of complicated functions.

Example

Let f be a function which is differentiable at a then

$$P_1(x) := f(a) + f'(a)(x - a)$$

The function $P_1(x)$ is a linear polynomial which approximates $f(x)$ at values close to a.

Definition

Suppose that f is n-times differentiable at a. Then the nth Taylor polynomial for f about a is defined by

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n.$$

In sigma notation, the Taylor polynomial P_n can be written as

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k.$$

Example

Find the nth Taylor polynomial of $f(x) = \sin x$ about 0.

$$f(x) = \sin x$$

$$f'(x) = \cos x$$

$$f''(x) = -\sin x$$

$$f^{(3)}(x) = -\cos x$$

$$f^{(4)}(x) = \sin x$$

$$\dots$$

If n is even, then the function is 0. If n is odd, then

$$P_n(x) = P_{2N+1}(x) = 1 \cdot x + \frac{-1}{3!}x^3 + \frac{1}{5!}x^5 + \cdots + \frac{(-1)^n}{(2N+1)!}x^{2N+1}$$

Example

Find the nth Taylor polynomial of $f(x) = e^x$ about 0.

$$P_n(x) = 1 + 1 \cdot (x - 0) + \frac{1}{2!}(x - 0)^2 + \cdots + \frac{1}{n!}(x - 0)^n$$

$$= 1 + x + \frac{1}{2!}x^2 + \cdots + \frac{1}{n!}x^n$$

Taylor's Theorem

Suppose that f has $n + 1$ continuous derivatives on an open interval I containing a. Then for each $x \in I$,

$$f(x) = P_n(x) + R_{n+1}(x),$$

where $P_n(x)$ is the nth Taylor polynomial about a, given by

$$P_n(x) = f(a) + \frac{f'(a)}{1!}(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n$$

and $R_{n+1}(x)$ is the remainder given in integral form as

$$R_{n+1}(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x - t)^n dt$$

or in Lagrange form as

$$R_{n+1}(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1}$$

for some real number c between a and x.

7.1 Sequences of real numbers

A sequence is simply a function whose domain is a subset of the natural numbers with co-domain the real numbers

- We usually use a and b rather than f and g to denote a sequence
- The value of the sequence at n is usually written as a_n instead of $f(n)$
- Thus the following notations indicate sequences: $\{a_n\}_{n=0}^{\infty}$ or $\{a_n\}$

7.1.1 Monotonic and bounded sequences

Let a_n be a non-decreasing or increasing sequence that is bounded above, ie.

$$a_0 \le a_1 \le \cdots \le a_{n-1} \le a_n \le \cdots \le K$$

or

$$a_0 < a_1 < \cdots < a_{n-1} < a_n < \cdots \le K$$

Then a_n converges to some limit $L \le K$.

Example

Let $f(x) = \sqrt{1 + 3x}$ and suppose that $a_1 = 4$ and $a_{n+1} = f(a_n)$ for all $n \geq 1$.

1. Prove that $a_{n+1} < a_n$ for all $n \geq 1$.

   Proving for $n = 1$

   $$a_2 = f(a_1) = \sqrt{1 + 3a_1} = \sqrt{13} < a_1 = 4$$

   Therefore, is true for $n = 1$. Suppose is true for $n = N$

   $$a_{N+2} = f(a_{N+1}) = \sqrt{1 + 3a_{N+1}} < \sqrt{1 + 3a_N} = a_{N+1}$$

   Therefore, is true for all n

2. Explain why there is a number L such that $a_n \to L$ as $n \to \infty$.

   $\{a_n\}$ is decreasing

   $$a_n \geq 0 \Rightarrow L = \lim_{n\to\infty} a_n \text{ exists}$$

3. Find the value of L.

   $$L \leftarrow a_{n+1} = f(a_n) = \sqrt{1 + 3a_n} \to \sqrt{1 + 3L}$$

   $$L = \sqrt{1 + 3L} \quad (L \geq 0)$$

   $$L^2 = 1 + 3L \Rightarrow L = \frac{3 \pm \sqrt{13}}{2} \quad \text{and} \quad L \geq 0$$

   So, $L = \frac{3+13}{2}$

Definition

A sequence $\{a_n\}$ of real numbers is said to be bounded above if there exists $M \in \mathbb{R}$ such that

$$a_n \leq M \quad \text{for all } n \in \mathbb{N}.$$

M is called an upper bound for $\{a_n\}$.

Every non-empty set of real numbers that has an upper bound has a least upper bound (or supremum). The least upper bound of a sequence $\{a_n\}$, if it exists, is denoted by $\sup_{n\in\mathbb{N}} a_n$ or $\sup\{a_n : n \in \mathbb{N}\}$

Example

Find the supremum of the sequence

$$\left\{\sin n + \frac{(-1)^n}{n} : n = 1, 2, \ldots\right\}$$

$$= \left\{\sin 1 + (-1), \sin 2 + \frac{1}{2}, \sin 3 + \frac{(-1)^3}{3}, \ldots\right\}$$

For $n \geq 3$:

$$\sin n + \frac{(-1)^n}{n} \leq 1 + \frac{1}{3} = 1.33333$$

Supremum is $\sin 2 + \frac{1}{2}$.

- If $M = \sup S$ then any number less than M is not an upper bound of S
- The supremum does not have to be attained in the set

Example

Let $S = \left\{4 - \frac{1}{n} : n = 1, 2, 3, \ldots\right\}$. Find $\sup S$ if it exists.

1. S is nonempty
2. S has an upper bound (as $4 - \frac{1}{n} \leq 4$), and so $\sup S$ exists
3. We claim $\sup S = 4$
   (a) 4 is an upper bound
   (b) For any $\epsilon > 0$, $4 - \epsilon$ cannot be an upper bound for S, that is

$$4 - \epsilon < 3 - \frac{1}{n_0} \text{for some} n_0$$

$$\Leftrightarrow n_0 > \frac{1}{\epsilon}$$

Chapter 8

Infinite series

- An infinite series is the sum of an infinite sequence:

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

- How to add together infinitely many numbers, ie. the limiting behaviour of the sum
- Eg.

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = 1$$

Let $\{a_k\}$ be a given sequence.

1. An expression of the form $\sum_{k=1}^{\infty} a_k$ or

$$a_1 + a_2 + a_3 + \cdots + a_k + \cdots$$

   is an infinite series.

2. The number a_k is called the kth term of the series $\sum_{k=1}^{\infty} a_k$
3. The sequence $\{s_n\}$ defined by

$$s_n = \sum_{k=1}^{n} a_k$$

Telescoping series

A type of infinite series which we can deal with is the telescoping series.

Example

Discuss the convergence of $\sum_{k=1}^{\infty} \frac{k}{(k+1)!}$.

Partial sum:

$$\begin{aligned} s_n = \sum_{k=1}^{n} a_k &= \sum_{k=1}^{n} \frac{k}{(k+1)!} \\ &= \sum_{k=1}^{n} \frac{(k+1)-1}{(k+1)!} \\ &= \sum_{k=1}^{n} \left[\frac{1}{k!} - \frac{1}{(k+1)!}\right] \\ &= \left[\frac{1}{1!} - \frac{1}{2!}\right] + \left[\frac{1}{2!} - \frac{1}{3!}\right] + \cdots + \left[\frac{1}{n!} - \frac{1}{(n+1)!}\right] \\ &= 1 - \frac{1}{(n+1)!} \to 1 \end{aligned}$$

So, $\sum_{k=1}^{\infty} a_k$ converges to 1.

$$a_k = b_k - b_{k+1}$$

a_k can be written as differences of successive terms from a sequence. If $\{b_k\}$ converges, say $b_k \to l$ as $k \to \infty$, then $s_N \to b_0 - l$ as $N \to \infty$.

Geometric series

Let $r \in \mathbb{R}$ and consider the series $\sum_{k=0}^{\infty} r^k$.

1. If $|r| < 1$ then the series converges, and $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$
2. If $|r| \geq 1$ then the series diverges.

Harmonic series

Example

Show that the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges.

$$\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k} + \cdots$$

The partial sum

$$\begin{aligned} s_n &= \sum_{k=1}^{n} \frac{1}{k} \\ &= 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \\ &\geq \int_1^n f(x)dx \text{ where } f(x) = \frac{1}{x} \\ &= [\ln x]_{x=1}^{x=n} = \ln n \to \infty \end{aligned}$$

Therefore $\int \frac{1}{k}$ diverges.

Example

Show that the series $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges

The partial sum

$$\begin{aligned} s_n = \sum_{k=1}^{n} \frac{1}{k^2} &= 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \\ &= 1 + \left[\frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2}\right] \\ &\leq 1 + \int_1^n f(x)dx \text{ where } f(x) = \frac{1}{x^2} \\ &= 1 + \int_1^n \frac{1}{x^2}dx \\ &= 2 - \frac{1}{n}. \end{aligned}$$

Since S is increasing and $s_n \leq 2 - \frac{1}{n} \leq 2$, the limit S exists. The limit is not 2 however, it is $\frac{\pi^2}{6}$

- $\sum_{k=0}^{\infty} a_k$ converges if and only if $\sum_{k=N}^{\infty} a_k$ converges. So, we sometimes write $\sum^{\infty} a_k$ if we are only after the convergence behaviour of the series.

- If $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ both converge and α is a constant then

$$\sum_{k=0}^{\infty}(a_k + b_k) = \sum_{k=0}^{\infty} a_k + \sum_{k=0}^{\infty} b_k$$

$$\sum_{k=0}^{\infty}(\alpha a_k) = \alpha \sum_{k=0}^{\infty} a_k$$

The kth term test for divergence

If a_k doesn't approach 0 as $k \to \infty$, then the series $\sum_{k=0}^{\infty} a_k$ diverges.

Proof: As $\sum_{k=0}^{\infty}$ converges, then $s_n = \sum_{k=1}^{n} a_k \to L$ (partial sum)

$$a_n = \sum_{k=1}^{n} a_k - \sum_{k=1}^{n-1} a_k = s_n - s_{n-1} \to 0 \text{ as } n \to \infty.$$

Example

Discuss the convergence of $\sum_{k=0}^{\infty} \frac{2k}{3k+1}$.

$$\lim_{k\to\infty} a_k = \lim_{k\to\infty} \frac{2}{3k+1} = \lim_{k\to\infty} \frac{2}{3+\frac{1}{k}} = \frac{2}{3} \neq 0.$$

Integral test

Suppose that $f(x)$ is a positive and decreasing function on $[1, \infty)$. Let $a_k = f(k)$ for $k = 1, 2, 3, \ldots$

1. If $\int_1^{\infty} f(x)dx$ converges, then $\sum_{k=1}^{\infty} a_k$ converges.
2. If $\int_1^{\infty} f(x)dx$ diverges, then $\sum_{k=1}^{\infty} a_k$ diverges.

Example

Discuss the convergence of $\sum_{k=2}^{\infty} \frac{1}{k \log k}$.

Let $f(x) = \frac{1}{x \log(x)}$ for $x \geq 2$. f takes positive values on $[2, +\infty)$. $f(x)$ is decreasing on $[2, +\infty)$.

$$\int_2^{\infty} f(x)dx = \lim_{A \to \infty} \int_2^A f(x)dx = \lim_{A \to \infty} \int_2^A \frac{1}{x \log(x)} dx.$$

Let $u = \log(x)$

$$= \lim_{A \to \infty} [\log(u)]_{u=\log 2}^{u=\log A}$$

Therefore $\sum_{k=2}^{\infty} a_k$ diverges by integral test.

The p-series

The series

$$\sum_{k=1}^{\infty} \frac{1}{k^p}$$

converges if $p > 1$ and diverges if $p \leq 1$.

Comparison test

Suppose $0 \leq a_k \leq b_k$ when k is sufficiently large (ie. $k \geq N$) for some $N \geq 0$

- If $\sum k = 0^{\infty} b_k$ converges, then $\sum k = 0^{\infty} a_k$ also converges
- If $\sum k = 0^{\infty} b_k$ diverges, then $\sum k = 0^{\infty} a_k$ also diverges

Suppose that $a_k > 0$, $a_k > 0$, and

$$\lim_{k \to \infty} \frac{a_k}{b_k} = K \geq 0.$$

- If $K \neq 0$ then

$$\sum^{\infty} a_k \text{ converges if and only if } \sum^{\infty} b_k \text{ converges.}$$

- If $K = 0$ then

$$\sum^{\infty} a_k \text{ converges if } \sum^{\infty} b_k \text{ converges}$$

$$\sum^{\infty} b_k \text{ diverges if } \sum^{\infty} a_k \text{ diverges.}$$

Ratio test

Suppose we have a sequence a_k of positive terms, and suppose

$$\frac{a_{k+1}}{a_k} \to L \text{ as } k \to \infty$$

Then

- $\sum_{k=0}^{\infty} a_k$ converges if $L < 1$
- $\sum_{k=0}^{\infty} a_k$ diverges if $L > 1$
- The test is inconclusive if $L = 1$

8.1 Absolute and conditional convergence

- The integral test, comparison test, and ratio test are for series of positive terms
- If the terms of a series are all negative, we simply multiply the series by -1 to obtain a series with positive terms
- What if a series has a mixture of positive and negative terms

The series $\sum_{k=1}^{\infty} a_k$ is said to be absolutely convergent if the series $\sum_{k=1}^{\infty} |a_k|$ is convergent.

The series $\sum_{k=1}^{\infty} a_k$ is said to be conditionally convergent if it converges but the series $\sum_{k=1}^{\infty} |a_k|$ diverges.

If $\{a_k\}$ is a sequence of positive numbers then the series

$$\sum_{k=0}^{\infty} (-1)^k a_k = a_0 - a_1 + a_2 - a_3 + \cdots$$

is called an alternating series

Suppose that $\{a_k\}$ is a sequence of real numbers satisfying the following properties

1. $a_k \geq 0$ for all k
2. $a_k \geq a_{k+1}$ for all k (ie. the sequence is non-increasing)
3. $\lim_{k \to \infty} a_k = 0$

Then the alternating series $\sum_{k=0}^{\infty} (-1)^k a_k$ converges.

8.1.1 Riemann

Let $\sum_{k=0}^{\infty} a_k$ be an infinite series

1. If the series is absolutely convergent, then every rearrangement of the series is absolutely convergent and has the same limit

2. If the series is conditionally convergent, then given any real number L there is a rearrangement that converges to L. There is also a rearrangement that diverges to ∞ and another rearrangement that diverges to $-\infty$.

8.1.2 Leibniz's test (alternating series tests)

Suppose that $\{a_k\}$ is a sequence of real numbers satisfying the following properties:

1. $a_k \geq 0$ for all k
2. $a_k \geq a_{k+1}$ for all k (ie. the sequence is non-increasing)
3. $\lim_{k\to\infty} a_k = 0$

Then the alternating series $\sum_{k=0}^{\infty}(-1)^k a_k$ converges.

Suppose that $\{a_n\}$ is a sequence of numbers satisfying the hypotheses of the Leibniz test. Let

$$s_n = \sum_{k=0}^{\infty}(-1)^k a_k \quad \text{and} \quad L = \sum_{k=0}^{\infty}(-1)^k a_k.$$

Then

$$|s_n - L| \leq a_{n+1} \ \forall n \in \mathbb{N}.$$

This result says that if you truncate the series after the nth term, the error in approximation will be less than the $(n+1)$st term.

8.1.3 Taylor's Series

Suppose that f has derivatives of all orders at a. Then the series

$$f(a) + \frac{f'(a)}{1!}(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

is called the Taylor Series (or Taylor expansion of f about a). When $a = 0$ the series

$$f(0) + \frac{f'(0)}{1!}x + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots = \sum_{k=0}^{\infty} \frac{f^{(k)}}{k!}x^k$$

is called the Maclaurin Series of f.

Let $\{a_k\}$ be a sequence of real numbers, and $a \in \mathbb{R}$. Then there exists an $R \in [0, \infty]$ (ie. R can be ∞) such that

- $\sum_{k=0}^{\infty} a_k(x-a)^k$ converges absolutely for $|x-a| < R$
- $\sum_{k=0}^{\infty} a_k(x-a)^k$ diverges for $|x-a| > R$

Chapter 9

Average, Arc-length, and Areas

9.1 Average

It is often desirable to find the average value of a quantity over a period of time, or space. Examples include the average temperature throughout the day, or the average height of trees in canopy.

If $f(x)$ is integrable on $[a, b]$ then the average $\overline{f}$ of $f(x)$ on the interval is

$$\overline{f} = \frac{\int_a^b f(x)dx}{b - a}.$$

Justification

Divide the interval into n subintervals of equal width $\Delta = \frac{b-a}{n}$ and let x_k denote a point in the kth subinterval then

$$\begin{aligned} \frac{\int_a^b f(x)dx}{b - a} &= \lim_{n\to\infty} \frac{\sum_{k=1}^n f(x_k)\Delta}{b - a} = \lim_{n\to\infty} \frac{\sum_{k=1}^n f(x_k)(b - a)}{(b - a)n} \\ &= \lim_{n\to\infty} \frac{1}{n}[f(x_1) + f(x_2) + \cdots + f(x_n)] \\ &= \overline{f} \end{aligned}$$

9.1.1 Mean Value Theorem for Integrals

Suppose that f is continuous on $[a, b]$. Then, there exists $c \in (a, b)$ such that

$$\int_a^b f(t)dt = f(x)(b - a).$$

9.2 Arc length

Consider a function $y = f(x)$.

Slice the length into small pieces, and approximate the length of each piece by the length l_k of the secant line joining (x_{k-1}, y_{k-1}) to (x_k, y_k).

$$l_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$$

$$l \approx \sum_{k=1}^{n} l_k = \sum_{k=1}^{n} \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} = \sum_{k=1}^{n} \sqrt{1 + \left(\frac{\Delta y_k}{\Delta x_k}\right)^2} \Delta x_k$$

Consider the limit $n \to \infty$ then $l = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\ dx$.

The arc length of the curve $y = f(x)$ between $x = a$and $x = b$ is

$$l = \int_a^b \sqrt{1 + (f'(x))^2}\ dx$$

(provided f is differentiable and the integral exists).

The arc length of a curve represented in parametric form $(x(t), y(t))$ between $t = a$ and $t = b$ is given by

$$l = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\ dt.$$

Suppose that a curve is described using polar coordinates by

$$r = f(\theta), \quad \theta_1 \le \theta \le \theta_2$$

The arc length of a polar curve $r = f(\theta)$ is given by

$$l = \int_{\theta_1}^{\theta_2} \sqrt{f(\theta)^2 + (f'(\theta))^2}\ d\theta = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\ d\theta.$$

9.3 Speed of a particle

Consider a particle moving in the plane along a trajectory $c(t) = (x(t), y(t))$

- The velocity of the particle is $v(t) = (x'(t), y'(t))$.
- Thus, at time t, the speed of the particle is the magnitude of the vector $v(t)$ which can be computed as

$$\sqrt{(x'(t))^2 + (y'(t))^2}$$

- From an arc-length perspective, the distance travelled by the particle from time t_0 to time t along the trajectory is

$$l(t) = \int_0^t \sqrt{(x'(u))^2 + (y'(u))^2}\ du$$

and the speed of the particle at time t is just the rate of change of distance with respect to time. Thus, which is

$$l' = \sqrt{(x(t))^2 + (y(t))^2}$$

Chapter 10

Surface Area

If the function $f(x) \geq 0$ is continuously differentiable on $[a, b]$, the area of the surface generated by revolving the curve $y = f(x)$ about the x-axis is

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\ dx = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2}\ dx.$$

Slice $[a, b]$ into n subintervals $[x_{k-1}, x_k]$ for $k = 1, \ldots, n$. On each subinterval we approximate the surface area by the area of frustum.